Getting Started with the BJT Beamer Class

A Quick Introduction and User Guide

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Outline

Introduction

What is the BJT Class?

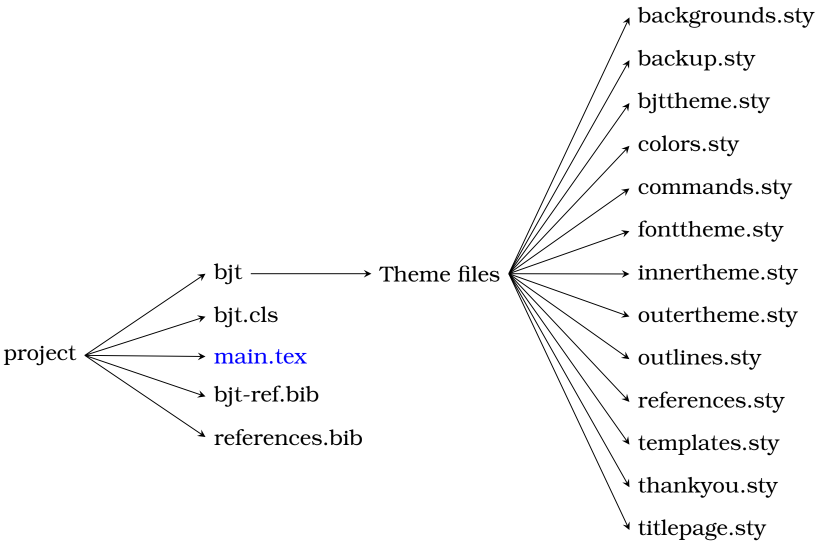
The `bjt` class is a personal Beamer class built on top of the **Metropolis** theme.

It provides

- A modern scientific presentation style
- Modular theme architecture
- Custom colors and typography
- Physics-oriented shortcuts
- Custom title and background pages
- Backup slide management
- Bibliography support
- Easy customization

Installation

Directory Structure



Using the Class

Creating a Presentation

Creating a presentation with the `bjt` class is straightforward.

```
\documentclass{bjt}

% Presentation information
\newcommand{\Title}{Title}                  % (EDIT HERE)
\newcommand{\Subtitle}{Subtitle}            % (EDIT HERE)
\newcommand{\AuthorName}{Name}              % (EDIT HERE)
\newcommand{\InstituteName}{University} % (EDIT HERE)
\newcommand{\PresentationDate}{\today}   % (EDIT HERE)

% Beamer metadata
\title{\Title}              % (DO NOT EDIT HERE)
\subtitle{\Subtitle}        % (DO NOT EDIT HERE)
\author{\AuthorName}        % (DO NOT EDIT HERE)
\institute{\InstituteName}  % (DO NOT EDIT HERE)
\date{\PresentationDate}    % (DO NOT EDIT HERE)

\begin{document}
\titlepage
\end{document}
```

The `bjt` class automatically loads the theme and the required packages, so no additional setup is necessary.

Backgrounds

Title Background

A custom title background can be added:

```
\titlebackground[0.25]{image.jpg}
\maketitleframe
```

The first argument controls opacity. The opacity argument is optional.

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Slide Background

Background images can also be applied globally:

```
\slidebackground[0.15]{wallpaper.jpg}
```

The first argument controls opacity. The opacity argument is optional.

Note

Using `\slidebackground{}` resets the previously applied background.

Typography

Design Philosophy

The class follows a modular structure.

Core Theme

Controls global appearance:

- Colors
- Fonts
- Inner theme
- Outer theme

Templates

Provides reusable components:

- Title page
- Backgrounds
- References
- Backup slides

Fonts

This theme uses

- Bookman as the text font
- EulerVM for mathematics
- Small-cap frame titles
- Serif font theme

Example:

$$E = mc^2$$

$$\int_0^\infty e^{-x^2}\,dx = \frac{\sqrt{\pi}}{2}$$

Blocks

Block Environments

Normal Block

This is the standard block.

Alert Block

Use this for warnings or important remarks.

Example Block

Useful for examples and exercises.

Lists

Itemized Lists

The itemize environment uses customized checkmarks.

- First item
- Second item
- Third item

Enumerations are unchanged.

1. One
2. Two
3. Three

Figures

FIGURES

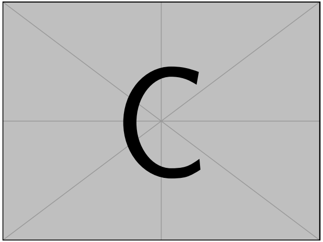

Figure 1: LaTeX logo

Captions are automatically numbered.

Tables

Tables

Parameter	Value	Unit
Mass	1.2	kg
Velocity	15	m/s
Time	12	s

Mathematics

Displayed Mathematics

The class loads the AMS packages automatically.

$$E = mc^2$$

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}$$

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

Customization

Changing Colors

Modify

```
bjt/colors.sty
```

Example:

```
\setbeamercolor{structure}{fg=purple}

\setbeamercolor{progress bar}{fg=orange,bg=gray}
```

Edit

```
bjt/fonttheme.sty
```

For example,

```
\RequirePackage{bookman}
\RequirePackage{eulervm}
```

Replace them with your preferred font packages.

Changing Blocks

Modify

```
bjt/innertheme.sty
```

For example,

```
\setbeamertemplate{blocks}[rounded][shadow=true]
```

or

```
\setbeamertemplate{itemize item}{\ensuremath{\checkmark}}
```

Changing the Footline

Modify

```
bjt/outertheme.sty
```

For example,

```
\setbeamertemplate{footline}[frame number]
```

You can also adjust the progress bar thickness.

References

References can be inserted automatically:

```
\referencesframe{references.bib}
```

Note

\referencesframe{references.bib} sets the bib file. `unsrt` is the default bibliography style.

or with a custom style:

```
\referencesframe[abbrvnat]{references.bib}
```

Summary

The BJT Beamer Class provides

- Modular theme architecture
- Modern Metropolis design
- Custom title pages
- Background control
- Physics-oriented commands
- Bibliography management
- Backup slide support
- Easy extension through packages

Designed for scientific presentations. Happy LaTeXing!

Questions?

Thank You!

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Backup Slides

Extra Derivation

This is a backup slide.

Equivalence of Mass and Energy

$$E = m c^2$$

Equivalence of Mass and Energy

$$\boxed{E} = m \; c^2$$

Energy

Equivalence of Mass and Energy

$$E = \boxed{m}\, c^2$$

Mass

Equivalence of Mass and Energy

$$E = m\boxed{c^2}$$

Speed of Light

Source Code

This whole project is available at this link
https://github.com/Baijayanta21/bjt-beamer-theme.